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Answer introduction

Tech Solutions Inc. is building a custom ERP system for a large manufacturing client. The system needs to tie together Inventory tracking (to ensure no shortage of materials), Supply chain operations (to keep production running smoothly), Employee scheduling (to optimize employee working hours), and Real-time reporting (so management can make fast decisions).

The ERP system must remain adaptable to future enhancements to incorporate additional features such as AI-powered predictive analytics. To ensure these requirements, the team is using the Object-Oriented Design methodology, and this essay analyses the application of OOD in the ERP project and evaluates whether Agile or Waterfall is the most suitable methodology.

The Application of Object-Oriented Design in the ERP Module

Object-Oriented Design is an approach that structures a system around interacting with software objects. According to Adenowo and Adenowo (2013), OOD combines data and processes into objects that communicate through methods. This makes it suitable for ERP systems because manufacturing environments contain many connected

components, including inventory management, suppliers, employees, and reporting services.

One significant advantage of OOD is scalability. The ERP system can be divided into independent classes and objects, allowing new features to be introduced without affecting the entire system. For example, AI-driven predictive analytics can later be added as a separate component while continuing to interact with inventory and reporting functions.

OOD also improves maintainability through encapsulation. Encapsulation protects internal data and exposes only the functionality required by other objects. This allows developers to make changes without disrupting the rest of the application. In a manufacturing environment where downtime can be costly, this contributes to system reliability.

Inheritance supports code reuse by allowing specialized classes to inherit features from more general classes. For example, managers and warehouse employees can share common employee characteristics while maintaining unique responsibilities.

Polymorphism further improves flexibility by allowing different objects to respond differently to the same action, making future enhancements easier to implement.

Another valuable OOD tool is UML modelling. Use case, class, sequence, and activity diagrams provide an easy to interpret visual representation of system interactions and requirements. UML assists developers in understanding business processes while improving communication with stakeholders and reducing misunderstandings during development.

Specific OOD techniques like Factory and Observer design patterns also strengthen the ERP solution. Factory patterns simplify the creation of system objects, while Observer patterns support real-time reporting by automatically updating related components when information changes. This is particularly beneficial in a manufacturing environment because managers can monitor inventory levels and production performance immediately, reducing the risk of operational delays. Both approaches improve software quality, reusability, and consistency.

Despite these advantages, OOD can introduce complexity during the design stage. Developers must carefully identify objects, relationships, and responsibilities. Poor design decisions can make the system difficult to maintain. However, for a large ERP solution requiring future expansion and integration, the long-term benefits of flexibility and modularity outweigh these challenges.

According to Meilani and Purnama (2023), object-oriented programming improves flexibility, simplifies maintenance, and makes applications easier to develop because functionality is organized into reusable classes and objects.

OOD in Agile versus Waterfall Methodologies

The Waterfall approach uses a sequential process where it is compulsory to complete each phase before the next phase begins. This approach provides clear documentation, structure, and predictable planning. However, Adenowo and Adenowo (2013) note that revisiting completed phases can be difficult and expensive.

For the ERP project, Waterfall presents several risks. Requirements may change as the client gains a better understanding of reporting needs, supply chain processes, and AI

integration opportunities. Because Waterfall depends on fixed requirements at the start of development, changes may increase costs and delay delivery. The six-month project timeline leaves limited opportunity for major redesigns.

The Agile approach supports incremental and iterative development. It suits ERP development because teams can develop, test and refine functionality through short iterative sprints. This allows developers to deliver working functionality earlier while receiving regular feedback from the client.

Agile also allows inventory management, employee scheduling, and reporting features to be developed independently and improved throughout the project. This reduces development risks and ensures that high-priority functionality is delivered within the available budget and timeframe. Although Agile requires continuous stakeholder involvement and effective project management, its flexibility makes it more suitable for this ERP project than Waterfall.

In conclusion, OOD provides an effective approach for developing Tech Solutions Inc's proposed ERP module. Through encapsulation, inheritance, polymorphism, UML modelling, and design patterns, OOD supports scalability, maintainability, reliability, and future adaptability. While Waterfall offers structure and strong documentation, its limited flexibility makes it less suitable for a complex ERP system with changing requirements. Agile combined with iterative OOD implementation is the better choice because it supports stakeholder feedback, continuous improvement, and efficient delivery within the project's six-month timeline. Therefore, Agile OOD offers the most practical and reliable solution for successfully delivering the ERP module.

Sources

Adenowo, A.A. & Adenowo, B.A. 2013. Software engineering methodologies: A review of the waterfall model and object-oriented approach. International Journal of Scientific & Engineering Research. Available at: <http://www.ijser.org/> [Accessed 19 August 2025].

Meilani, Y.I. & Purnama, J. 2023. Object Oriented Programming of Application Admission of New High School Students. SINKRON Journal. Available at: <https://jurnal.polgan.ac.id/index.php/sinkron/article/view/12077/1320> [Accessed 20 February 2026].